Module 05 • Machine Learning & AI

Machine Learning Pipeline & Soil Moisture Forecasting

IRRIGATEWISE Hydrogel-Based Smart Irrigation System Specification

Machine Learning Pipeline & Soil Moisture Forecasting

1. Mathematical Formulation & Intuitive Formula Breakdown

The Core Continuous Differential Equation

Soil moisture dynamics in an agricultural field amended with superabsorbent hydrogel polymers can be modeled physically as a continuous mass-balance differential equation:

> **Equation:** `∂θ/∂t = -(ET_o(Temp, Hum, Rad) / Z_r) + ((P + I) / Z_r) + Q_gel(θ, D_gel) - D_r(θ)`

Making Sense of the Math: The "Soil Water Bank Account" Analogy

If you find the calculus notation (∂θ/∂t, partial derivatives, Greek symbols) difficult to follow, think of soil moisture like a **water bank account**:

```
┌──────────────────────────────────────────────────────────────────────┐
│      SOIL WATER BALANCE BANK ACCOUNT       │
│                    │
│ [New Moisture] = [Old Moisture] - [Evaporation Loss] + [Water Influx]  │
│         + [Hydrogel Released] - [Drainage Leak]  │
└──────────────────────────────────────────────────────────────────────┘
```

In Plain Words:

> **Equation:** `Change in Soil Moisture = -Evaporation Loss (Sun & Wind) + Water Added (Rain + Pump) + Hydrogel Water Release (Sponge) - Gravity Drainage Leak`

In Discrete Hourly Formula Form (How the Code Computes It Each Hour):

> **Equation:** `θ_next hour = θ_current hour - Δθ_evap + Δθ_rain + Δθ_irrigation + Δθ_hydrogel - Δθ_drainage`

Component-by-Component Formula Breakdown

Here is what every single symbol and term means, broken down into simple, transparent formulas:

Term in Differential Equation	Simple Name	What It Represents in Plain English	Practical Discrete Formula	Code Implementation
∂θ/∂t	**Moisture Rate of Change**	How fast soil is getting wetter (+) or drier (-) per hour (%/hour).	(θ(t) - θ(t-1)) / Δt	`moisture_rate_of_change`
-(ET_o(Temp,Hum,Rad) / Z_r)	**Evapotranspiration Loss**	Water evaporated by heat/wind and sucked up by plants to cool themselves.	Δθ_evap = (ET_o / (Z_r × 10)) × K_c	`hourly_loss = et0 * et_mult * et_retention * 0.45`
+((P + I) / Z_r)	**Moisture Influx**	Water added to the field from Rainfall (P) and Irrigation pump (I).	Δθ_gain = ((P + I) / (Z_r × 10)) × η_absorption	`current_moisture += rainfall * 0.85 + irrigation * 1.5`
+Q_gel(θ, D_gel)	**Hydrogel Desorption Flux**	Micro-reservoirs of hydrogel releasing trapped moisture when soil dries.	Q_gel = κ_gel · D_gel · max(0, 1 - (θ / θ_FC))	`et_retention = 1.0 / (1.0 + 0.015 * dosage)`
-D_r(θ)	**Gravitational Drainage**	Water leaking straight down past roots into deep aquifers when oversaturated.	D_r = excess × K_drain × Damping	`drainage_loss = excess * drain_rate * drain_damping`

Detailed Term Definitions & Parameter Units:

1. **θ(t) (Soil Moisture Content):** Measured in percentage (%, e.g., 55.0%). Field Capacity for loam is ≈ 60%, and crop stress threshold is ≈ 45%.
2. **Z_r (Root Zone Depth):** The depth of soil where crop roots feed, measured in meters (m, e.g., 0.4 m for typical cereal/vegetable crops).
3. **ET_o (Reference Evapotranspiration):** Measured in mm/hour. Higher temperature and wind speed increase ET_o; higher relative humidity decreases ET_o.
4. **P (Precipitation):** Natural rainfall depth in millimeters (mm).

5. **I (Irrigation Pulse):** Water delivered by drip lateral lines in millimeters (mm or L/m^2).
6. **Q_gel (Hydrogel Buffer Flux):** Rate at which swollen hydrogel polymer beads (D_{gel} = 50 g/m^2) desorb moisture back into root micropores when soil matric suction exceeds polymer osmotic potential.
7. **D_r (Deep Drainage):** Occurs only when soil moisture exceeds Field Capacity ($\theta > \theta_{FC}$). Hydrogel physically swells and dampens this loss by up to 60%.

Step-by-Step Numerical Walkthrough: 1-Hour Practical Example

To see how the numbers work in practice, let's calculate what happens over a hot 1-hour midday period (13:00 to 14:00, Temperature = 34°C, Humidity = 45%, Wind = 14 km/h, No rain, Pump OFF):

Scenario A: Hydrogel-Amended Field (50 g/m^2)

- **Starting Moisture ($\theta(t)$):** 60.0%
- **Evaporative Demand ($\Delta\theta_{evap}$):** -1.6%
- **Hydrogel Slow-Release Cushion (Q_{gel}):** +0.7% (polymer feeds roots moisture)
- **Rainfall / Irrigation Influx ($\Delta\theta_{gain}$):** 0.0%
- **Deep Drainage Loss (D_r):** 0.0% (soil is not oversaturated)
- **Net Calculation:**

> **Equation:** `θ_t+1 = 60.0% - 1.6% + 0.7% = \mathbf59.1% (Net 1-hour loss: -0.9%)`

Scenario B: Control Field (NO Hydrogel, 0 g/m^2)

- **Starting Moisture ($\theta(t)$):** 60.0%
- **Evaporative Demand ($\Delta\theta_{evap}$):** -1.6%
- **Hydrogel Slow-Release Cushion (Q_{gel}):** 0.0% (no polymer cushion)
- **Rainfall / Irrigation Influx ($\Delta\theta_{gain}$):** 0.0%
- **Deep Drainage Loss (D_r):** 0.0%
- **Net Calculation:**

> **Equation:** `θ_t+1 = 60.0% - 1.6% + 0.0% = \mathbf58.4% (Net 1-hour loss: -1.6%)`
>
> **Key Takeaway:** The hydrogel-amended soil dries **nearly 44% slower** under identical atmospheric stress, providing extended hydration buffer time and allowing irrigation cycles to be spaced much further apart!

The Multi-Horizon Forecasting Objective

Rather than estimating current moisture (which is directly measured by capacitive sensors at t=0), the ML task is to predict the **future moisture trajectory** at three distinct operational horizons: 1. **t + 1h**: Immediate short-term trend verification and sensor drift screening. 2. **t + 3h**: Primary operational decision window (triggering pump before root zone breaches critical stress). 3. **t + 6h**: Strategic diurnal planning horizon (evaluating whether midday solar peaks or evening cooling will alter depletion rates).

Given an observation feature vector \mathbfX_t, we train a multi-output regressor f_\Theta:

> **Equation:** `\hat\mathbfY_t = [θ̂t+1, θ̂t+3, θ̂t+6] = f_\Theta(\mathbfX_t)`

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2. Feature Engineering Pipeline

Raw sensor time-series undergo deterministic feature engineering implemented in `backend/ml/feature_engineering.py`:

```
Raw Sensor Stream (t) ──► [ Lag Generator ] ──────► θ(t-1), θ(t-2), θ(t-3)
      ──► [ Rolling Statistics ] ──► Mean_3h, Mean_6h, Std_3h, Std_6h
      ──► [ Diurnal Encodings ] ──► sin(2π h/24), cos(2π h/24)
      ──► [ Interaction Terms ] ──► Hydrogel * Temp, Hydrogel * Moisture
```

Engineered Feature Specification:

1. **Current Telemetry:** - `soil_moisture`: Instantaneous moisture content (%). - `temperature`: Ambient temperature (°C). - `humidity`: Ambient relative humidity (%). - `wind_speed`: Ambient wind speed (km/h). - `rainfall`: Precipitation depth (mm).
2. **Autoregressive Temporal Lags:** - `moisture_lag_1`: θ(t - 1) - `moisture_lag_2`: θ(t - 2) - `moisture_lag_3`: θ(t - 3) - `temp_lag_1`, `humidity_lag_1`: Short-term weather trajectory indicators. - Captures the immediate rate of drying or post-irrigation infiltration curve.
3. **Rolling Aggregate Statistics:** - `moisture_rolling_mean_3`: 3-hour moving average smoothing sensor micro-noise. - `moisture_rolling_mean_6`: 6-hour moving average capturing macro drying trend. - `moisture_rolling_std_3`: 3-hour standard deviation detecting infiltration events or sensor instability. - `moisture_rolling_std_6`: 6-hour standard deviation.
4. **Rate of Change & Evaporative Stress:** - `moisture_rate_of_change`: Instantaneous hourly delta θ(t) - θ(t-1). - `evap_stress_index`: Compound index Temperature × (1 - Humidity/100).
5. **Cyclical Diurnal Transformers:** - `hour_sin`: sin(2π · hour / 24) - `hour_cos`: cos(2π · hour / 24) - Eliminates the artificial discontinuity between hour 23:00 and 00:00, mapping solar heating smoothly.
6. **Hydrogel Specific Interaction Terms:** - `hydrogel_present`: Binary indicator (1 = Plot with Hydrogel, 0 = Control). - `hydrogel_dosage`: Application rate (g/m^2, baseline 50 g/m^2). - `hours_since_irrigation`: Cumulative drying duration since last watering pulse.

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3. Training & Model Comparison Methodology

Data Split Strategy (180-Day Dataset)

To prevent temporal data leakage, we employ a strict **Chronological Split** (not random k-fold) across the 180-day, N = 8,628 record physics-informed dataset: - **Training Set:** First 70% of chronological observations (N = 6,039). - **Validation Set:** Next 15% of chronological observations (N = 1,294). - **Test Set:** Final 15% of unseen chronological observations (N = 1,295).

Candidate Algorithms Evaluated:

1. **Ordinary Least Squares Multi-Output Linear Regression:** Baseline benchmark (R^2 = 0.3345, MAE = 2.95%).
2. **Random Forest Regressor:** Non-linear ensemble of 100 decorrelated decision trees (R^2 = 0.7013, MAE = 1.231%).
3. **Multi-Output Gradient Boosting Regressor (Active Production):** Sequential gradient boosted trees minimizing mean squared error residuals across all horizons (R^2 = 0.7312, MAE = 1.197%).

Multi-Horizon Production Performance (Gradient Boosting):

- **t+1h Forecast:** MAE = 0.473%, RMSE = 1.131%, R^2 = 0.9479
- **t+3h Forecast:** MAE = 1.198%, RMSE = 2.591%, R^2 = 0.7269
- **t+6h Forecast:** MAE = 1.919%, RMSE = 3.442%, R^2 = 0.5187

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4. Model Registry & Version Control

All trained models and their hyperparameters are tracked using `backend/ml/model_registry.py` and synchronized with the PostgreSQL `ml_model_versions` table:

```
SELECT id, model_name, version_tag, mae, rmse, r2_score, is_active, trained_at
FROM ml_model_versions;
```

When new training runs complete, the registry checks if the new candidate exceeds the active model's R^2 on the validation set before flipping the `is_active` flag, guaranteeing zero-downtime atomic model updates.